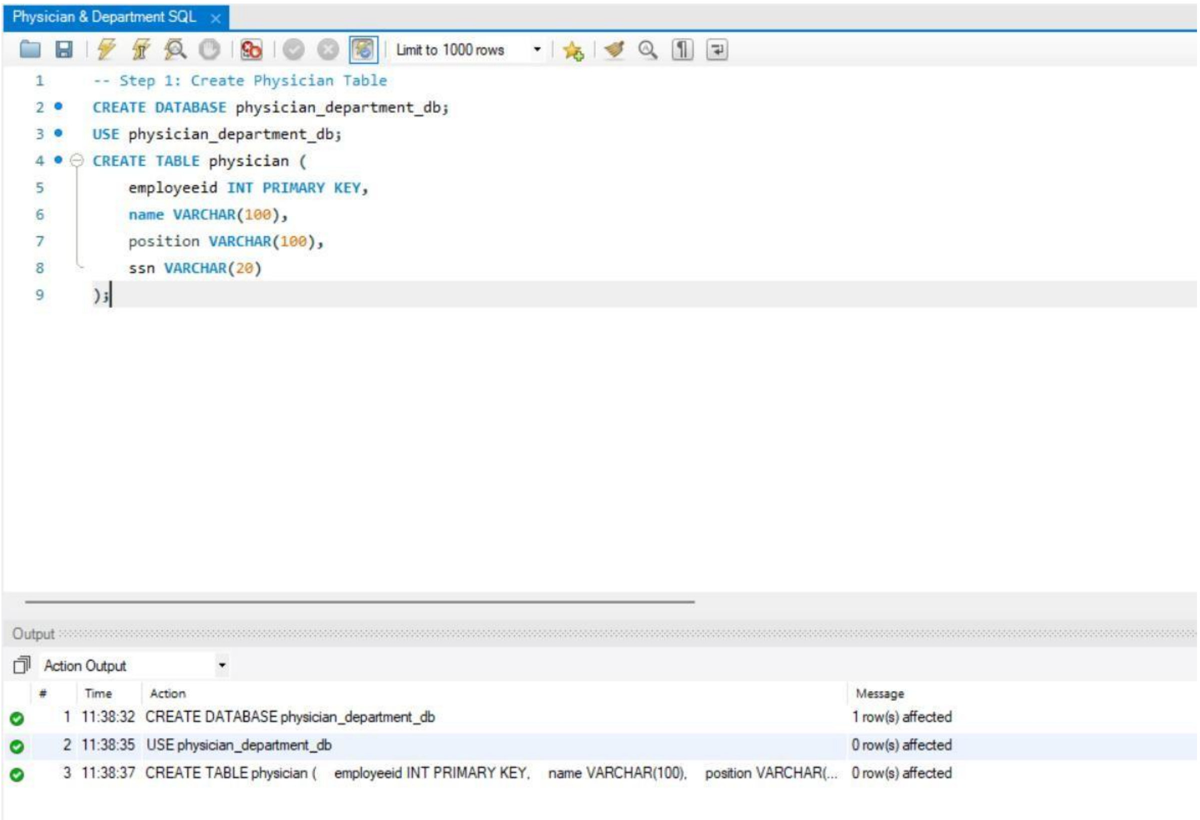
PHYSICIAN & DEPARTMENT SQL

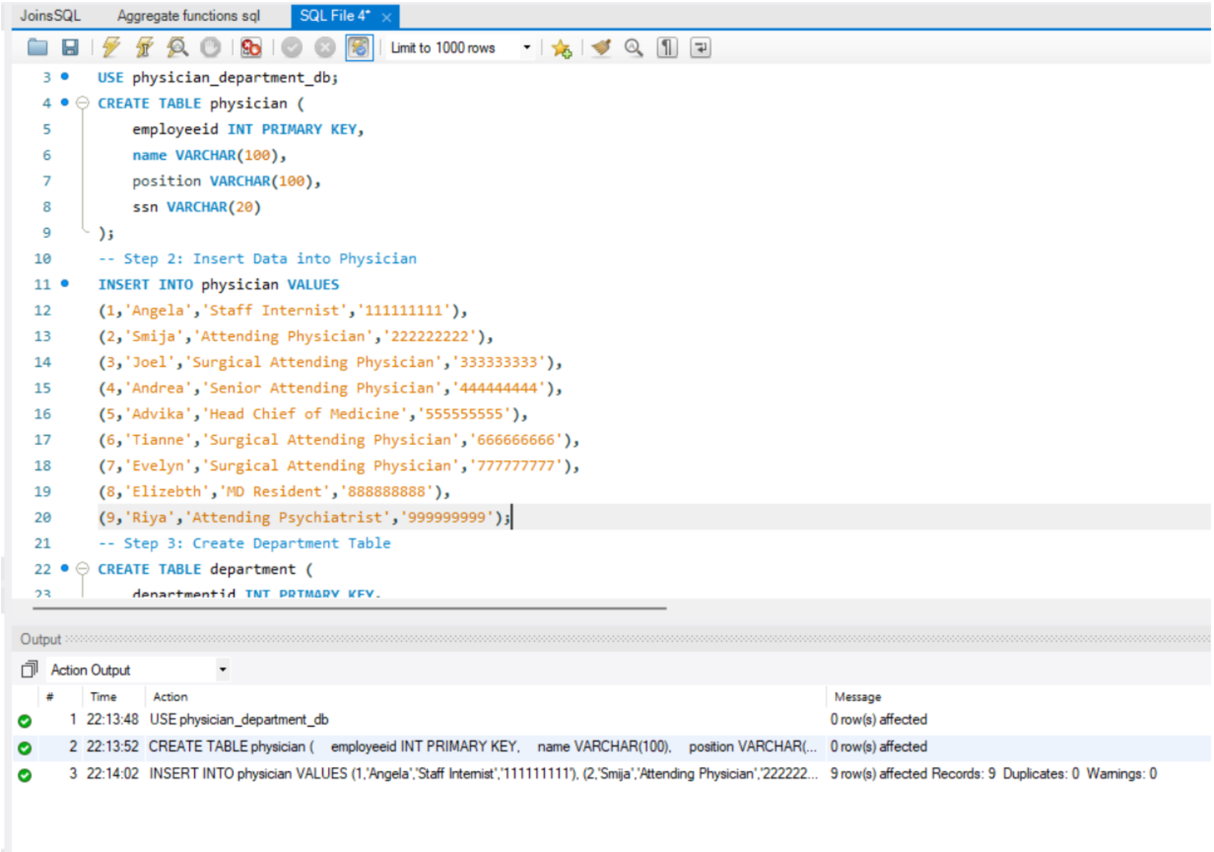
Angela Jose (2520030365)

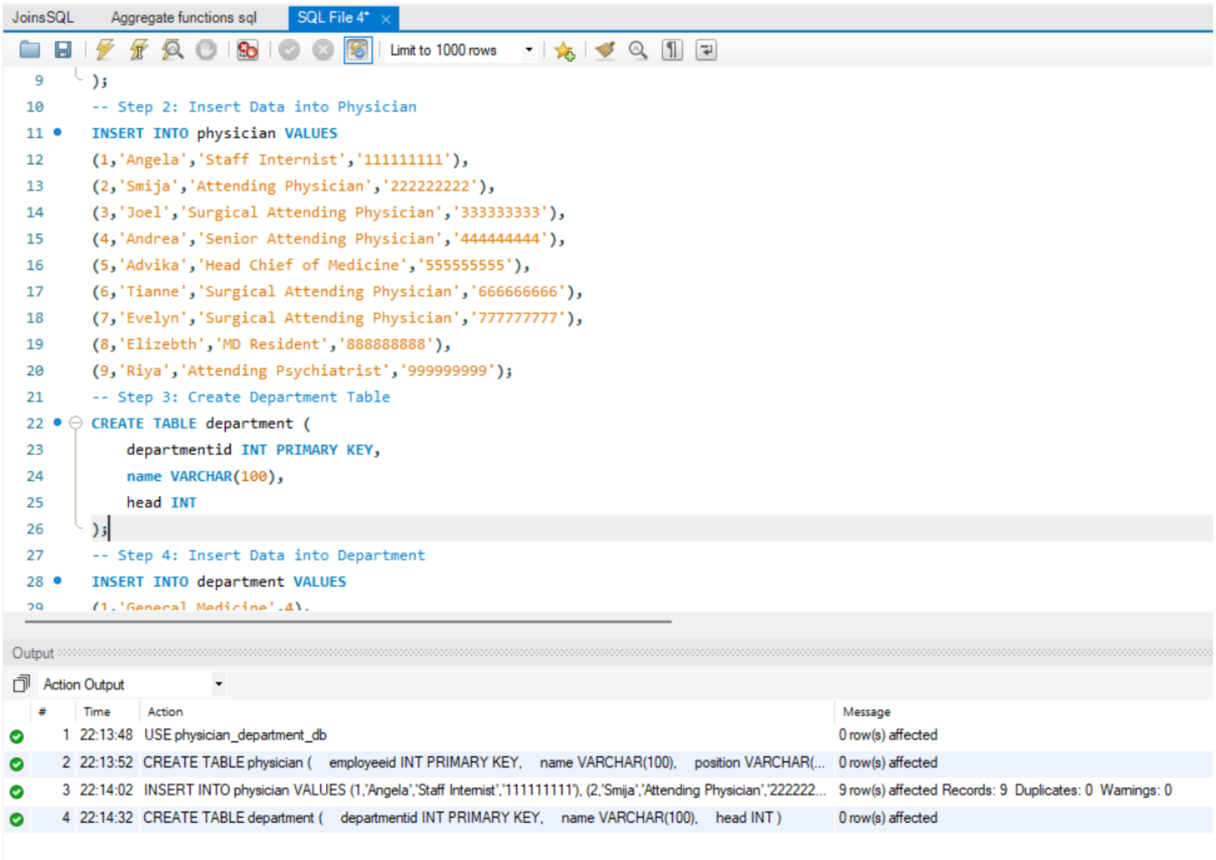
Question :

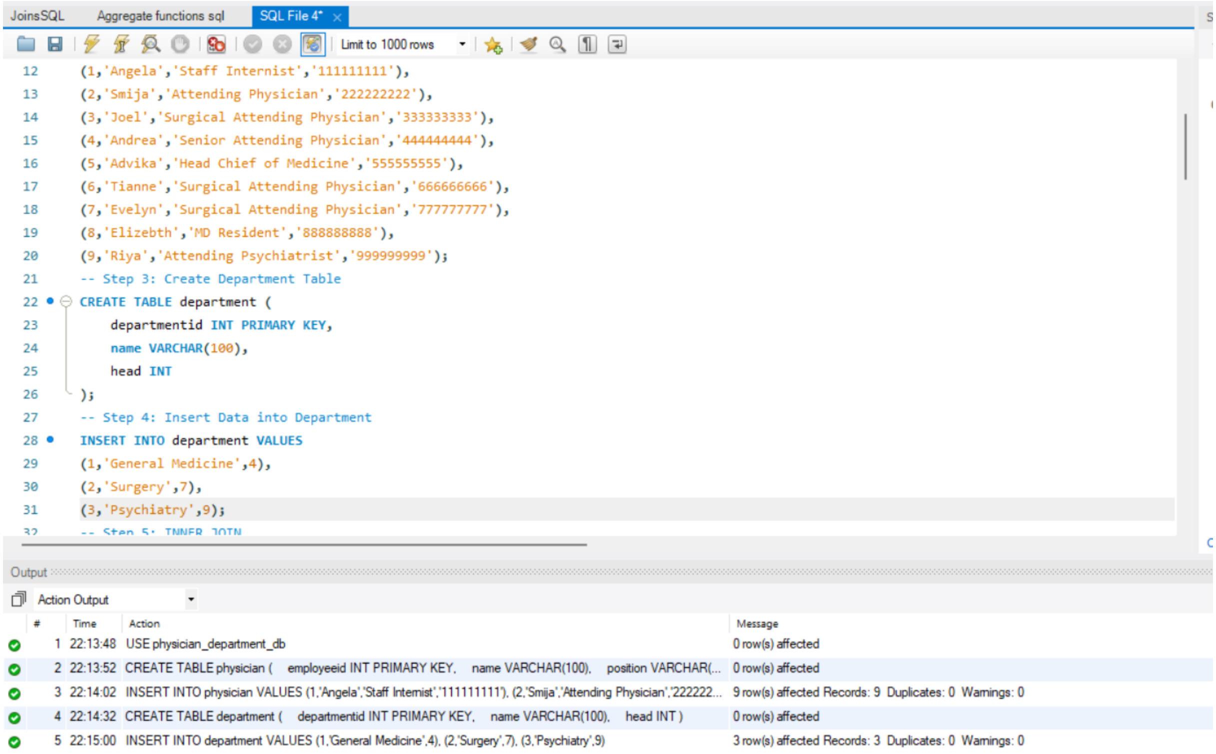
Perform SQL operations on Physician and Department tables using INNER JOIN, WHERE, LIKE, NOT LIKE, IN, NOT IN, EXISTS, GROUP BY, HAVING, ORDER BY, subqueries, COUNT(), COUNT(DISTINCT), MIN(), and MAX() to retrieve and analyse physician and department information.

Step 1 : Create Physician Database and Table

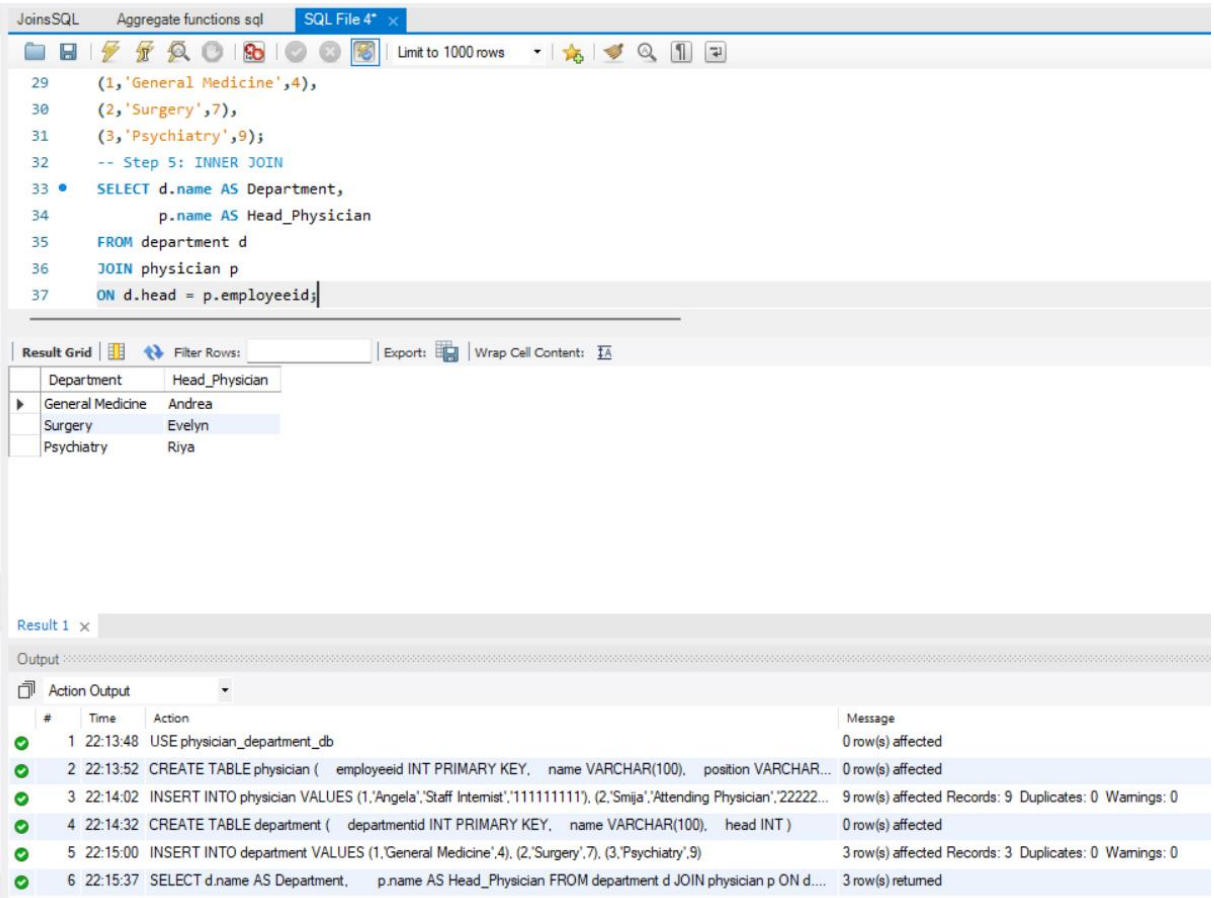



JoinsSQL
Aggregate functions sql
SQL File 4*
Limit to 1000 rows
USE physician_department_db;
CREATE TABLE physician (
employeeid INT PRIMARY KEY,
name VARCHAR(100),
position VARCHAR(100),
ssn VARCHAR(20)
);
-- Step 2: Insert Data into Physician
INSERT INTO physician VALUES
(1,'Angela','Staff Internist','111111111'),
(2,'Smija','Attending Physician','222222222'),
(3,'Joel','Surgical Attending Physician','333333333'),
(4,'Andrea','Senior Attending Physician','444444444'),
(5,'Advika','Head Chief of Medicine','555555555'),
(6,'Tianne','Surgical Attending Physician','666666666'),
(7,'Evelyn','Surgical Attending Physician','777777777'),
(8,'Elizebth','MD Resident','888888888'),
(9,'Riya','Attending Psychiatrist','999999999');
-- Step 3: Create Department Table
CREATE TABLE department (
Output
Action Output
Time Action Message
1 22:13:48 USE physician_department_db 0 row(s) affected
2 22:13:52 CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(... 0 row(s) affected
3 22:14:02 INSERT INTO physician VALUES (1,'Angela','Staff Internist','111111111'), (2,'Smija','Attending Physician','222222... 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0

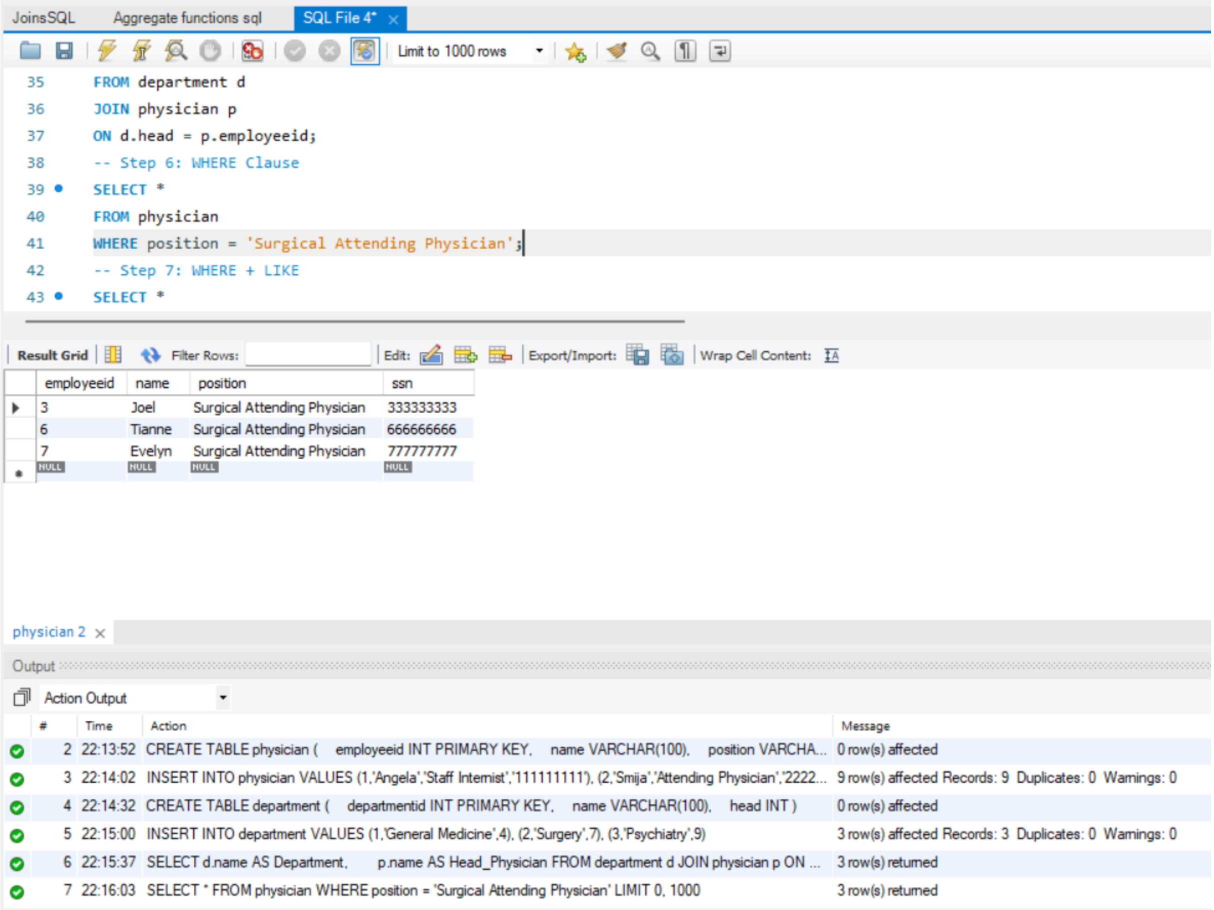

JoinsSQL
Aggregate functions sql
SQL File 4*
Limit to 1000 rows
);
-- Step 2: Insert Data into Physician
INSERT INTO physician VALUES
(1,'Angela','Staff Internist','111111111'),
(2,'Smija','Attending Physician','222222222'),
(3,'Joel','Surgical Attending Physician','333333333'),
(4,'Andrea','Senior Attending Physician','444444444'),
(5,'Advika','Head Chief of Medicine','555555555'),
(6,'Tianne','Surgical Attending Physician','666666666'),
(7,'Evelyn','Surgical Attending Physician','777777777'),
(8,'Elizebth','MD Resident','888888888'),
(9,'Riya','Attending Psychiatrist','999999999');
-- Step 3: Create Department Table
CREATE TABLE department (
departmentid INT PRIMARY KEY,
name VARCHAR(100),
head INT
);
-- Step 4: Insert Data into Department
INSERT INTO department VALUES
Output
Action Output
Time Action Message
1 22:13:48 USE physician_department_db 0 row(s) affected
2 22:13:52 CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(... 0 row(s) affected
3 22:14:02 INSERT INTO physician VALUES (1,'Angela','Staff Internist','111111111'), (2,'Smija','Attending Physician','222222... 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
4 22:14:32 CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT) 0 row(s) affected


JoinsSQL
Aggregate functions sql
SQL File 4*
Limit to 1000 rows
(1,'Angela','Staff Internist','111111111'),
(2,'Smija','Attending Physician','222222222'),
(3,'Joel','Surgical Attending Physician','333333333'),
(4,'Andrea','Senior Attending Physician','444444444'),
(5,'Advika','Head Chief of Medicine','555555555'),
(6,'Tianne','Surgical Attending Physician','666666666'),
(7,'Evelyn','Surgical Attending Physician','777777777'),
(8,'Elizebth','MD Resident','888888888'),
(9,'Riya','Attending Psychiatrist','999999999');
-- Step 3: Create Department Table
CREATE TABLE department (
departmentid INT PRIMARY KEY,
name VARCHAR(100),
head INT
);
-- Step 4: Insert Data into Department
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
Output
Action Output
Time Action Message
1 22:13:48 USE physician_department_db 0 row(s) affected
2 22:13:52 CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(... 0 row(s) affected
3 22:14:02 INSERT INTO physician VALUES (1,'Angela','Staff Internist','111111111'), (2,'Smija','Attending Physician','22222... 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
4 22:14:32 CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT) 0 row(s) affected
5 22:15:00 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

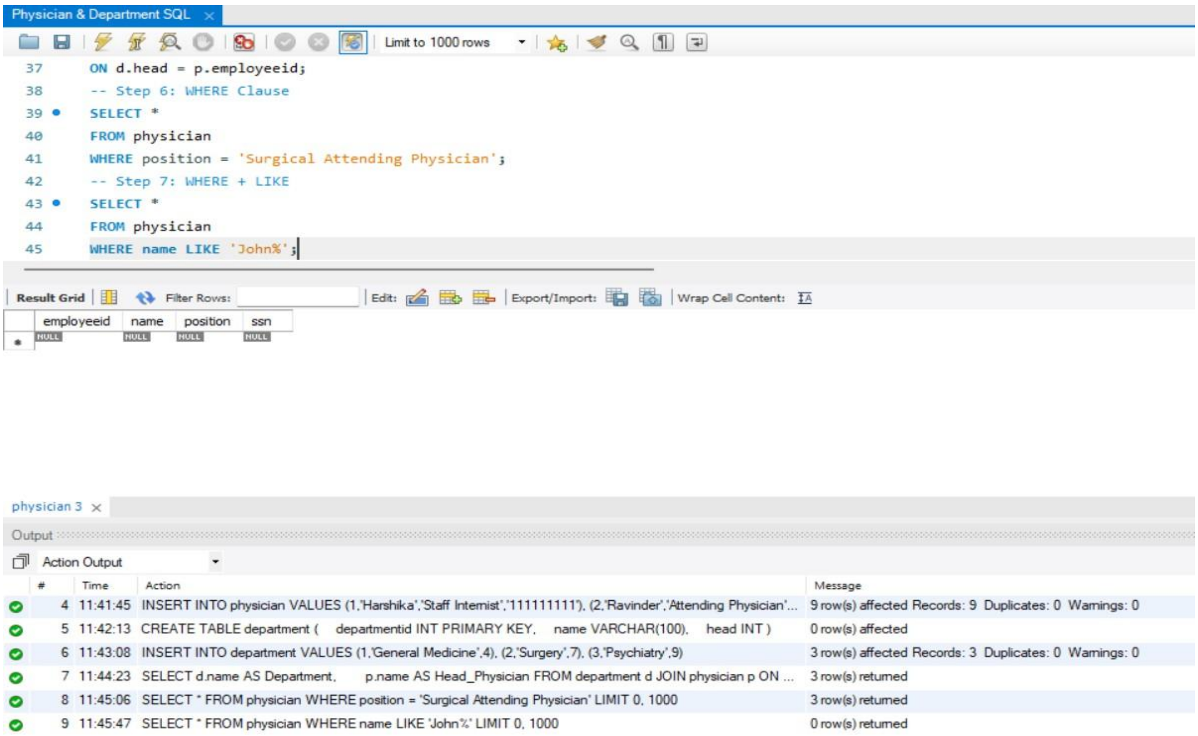
Step 2 : JOIN


JoinsSQL
Aggregate functions sql
SQL File 4*
Limit to 1000 rows
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
-- Step 5: INNER JOIN
SELECT d.name AS Department,
p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Department Head_Physician
General Medicine Andrea
Surgery Evelyn
Psychiatry Riya
Result 1
Output
Action Output
Time Action Message
1 22:13:48 USE physician_department_db 0 row(s) affected
2 22:13:52 CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR... 0 row(s) affected
3 22:14:02 INSERT INTO physician VALUES (1,'Angela','Staff Internist','111111111'), (2,'Smija','Attending Physician','22222... 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
4 22:14:32 CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT) 0 row(s) affected
5 22:15:00 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6 22:15:37 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.... 3 row(s) returned

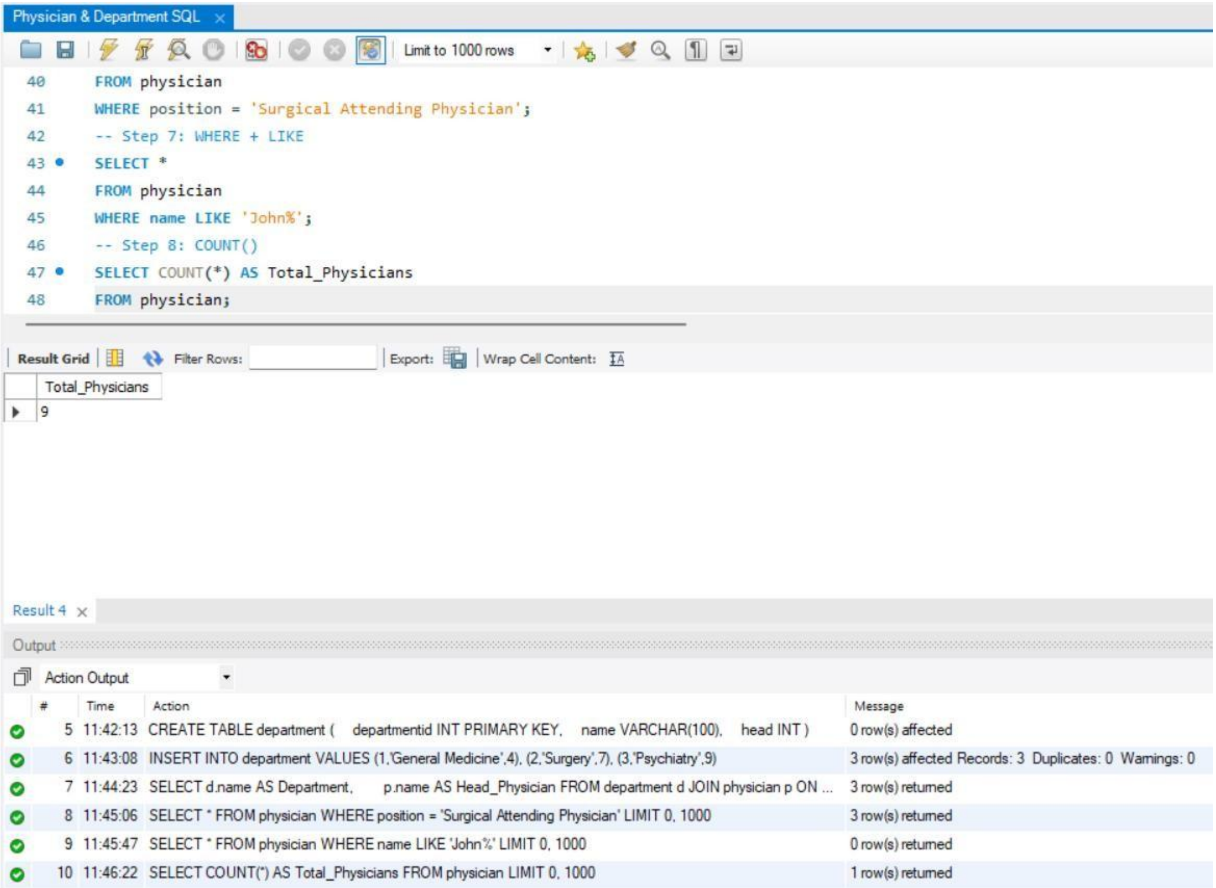
Step 3 : WHERE

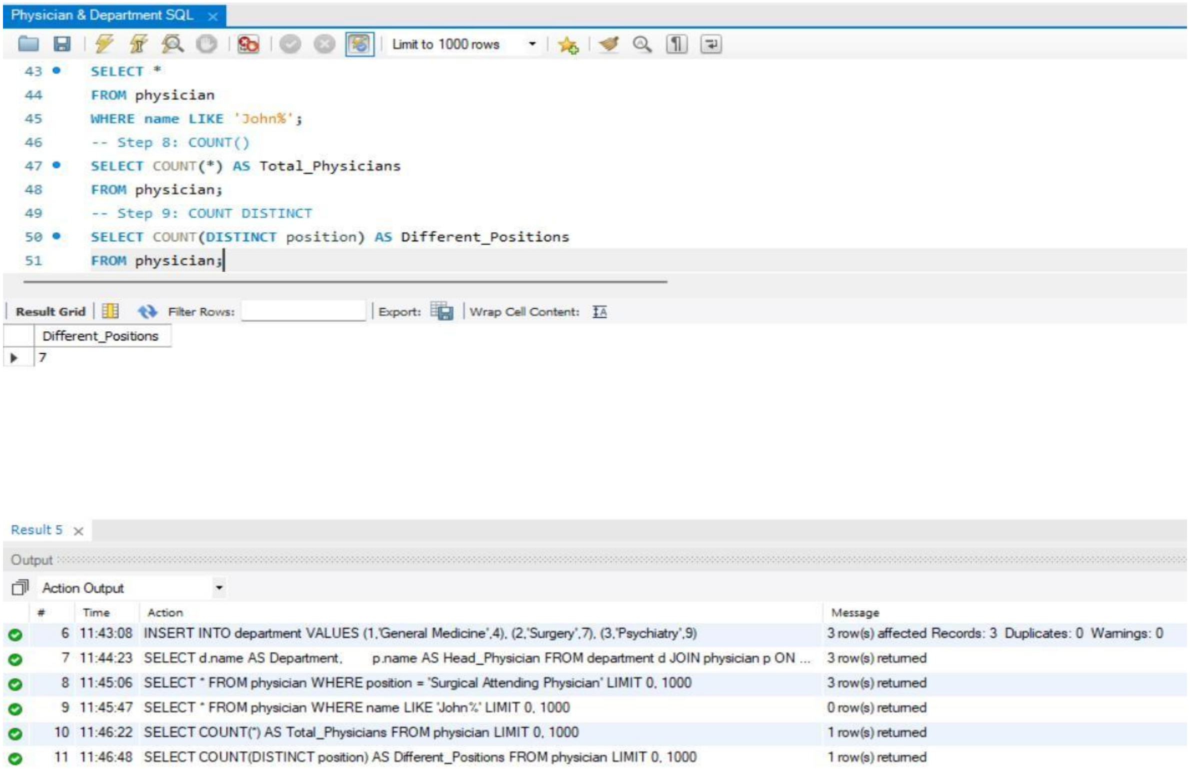


Step 4 : LIKE



Step 5 : COUNT


Physician & Department SQL
Limit to 1000 rows
FROM physician
WHERE position = 'Surgical Attending Physician';
-- Step 7: WHERE + LIKE
SELECT *
FROM physician
WHERE name LIKE 'John%';
-- Step 8: COUNT()
SELECT COUNT(*) AS Total_Physicians
FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Total_Physicians
9
Result 4
Output
Action Output
Time Action Message
5 11:42:13 CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT) 0 row(s) affected
6 11:43:08 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7 11:44:23 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned
8 11:45:06 SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned
9 11:45:47 SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000 0 row(s) returned
10 11:46:22 SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000 1 row(s) returned


Physician & Department SQL
Limit to 1000 rows
SELECT *
FROM physician
WHERE name LIKE 'John%';
-- Step 8: COUNT()
SELECT COUNT(*) AS Total_Physicians
FROM physician;
-- Step 9: COUNT DISTINCT
SELECT COUNT(DISTINCT position) AS Different_Positions
FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Different_Positions
7
Result 5
Output
Action Output
Time Action Message
6 11:43:08 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7 11:44:23 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned
8 11:45:06 SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned
9 11:45:47 SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000 0 row(s) returned
10 11:46:22 SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000 1 row(s) returned
11 11:46:48 SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000 1 row(s) returned

Step 6 : GROUP BY

Physician & Department SQL

```sql
FROM physician;
-- Step 9: COUNT DISTINCT
SELECT COUNT(DISTINCT position) AS Different_Positions
FROM physician;
-- Step 10: GROUP BY
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position;
```

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 6

Output

Action Output

#	Time	Action	Message
7	11:44:23	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned

Physician & Department SQL

```sql
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position;
-- Step 11: GROUP BY + HAVING
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

position	Total_Employees
Surgical Attending Physician	3

Result 7

Output

Action Output

#	Time	Action	Message
8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

Step 7: ORDER BY

JoinsSQL | Aggregate functions sql | SQL File 4*

Limit to 1000 rows

```sql
GROUP BY position
HAVING COUNT(*) > 1;
-- Step 12: ORDER BY ASC
SELECT *
FROM physician
ORDER BY name ASC;
-- Step 13: ORDER BY DESC
SELECT *
FROM physician
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employeeid	name	position	ssn
5	Advika	Head Chief of Medicine	555555555
4	Andrea	Senior Attending Physician	444444444
1	Angela	Staff Internist	111111111
8	Elizebth	MD Resident	888888888
7	Evelyn	Surgical Attending Physician	777777777
3	Joel	Surgical Attending Physician	333333333
9	Riya	Attending Psychiatrist	999999999
2	Smija	Attending Physician	222222222
6	Tianne	Surgical Attending Physician	666666666
NULL	NULL	NULL	NULL

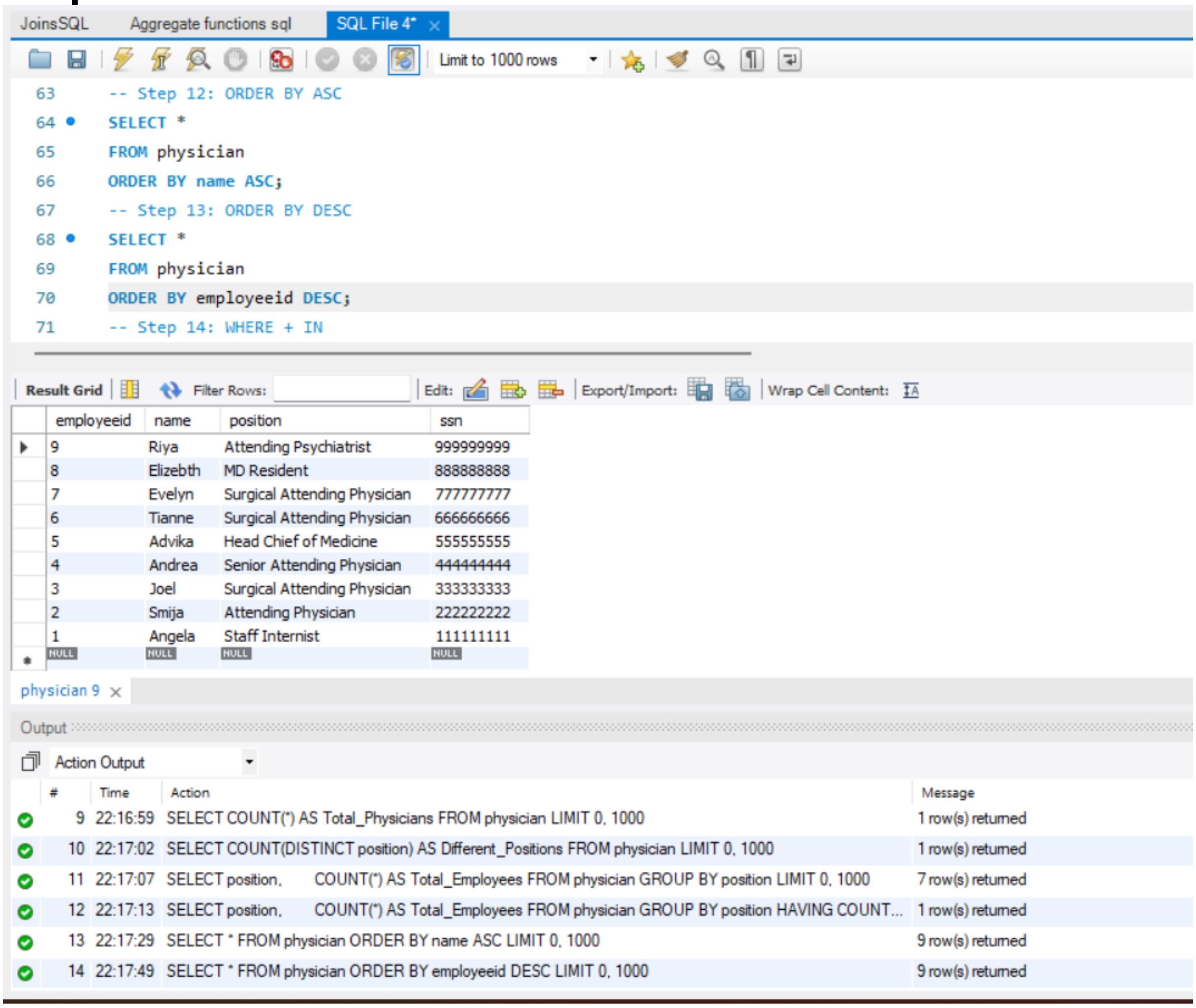
physician 8 ×

Output

Action Output

#	Time	Action	Message
8	22:16:54	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
9	22:16:59	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
10	22:17:02	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
11	22:17:07	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
12	22:17:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
13	22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned

Step 7: ORDER BY



JoinsSQL | Aggregate functions sql | SQL File 4*

Limit to 1000 rows

```sql
-- Step 12: ORDER BY ASC
SELECT *
FROM physician
ORDER BY name ASC;
-- Step 13: ORDER BY DESC
SELECT *
FROM physician
ORDER BY employeeid DESC;
-- Step 14: WHERE + IN
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employeeid	name	position	ssn
9	Riya	Attending Psychiatrist	999999999
8	Elizebth	MD Resident	888888888
7	Evelyn	Surgical Attending Physician	777777777
6	Tianne	Surgical Attending Physician	666666666
5	Advika	Head Chief of Medicine	555555555
4	Andrea	Senior Attending Physician	444444444
3	Joel	Surgical Attending Physician	333333333
2	Smija	Attending Physician	222222222
1	Angela	Staff Internist	111111111
NULL	NULL	NULL	NULL

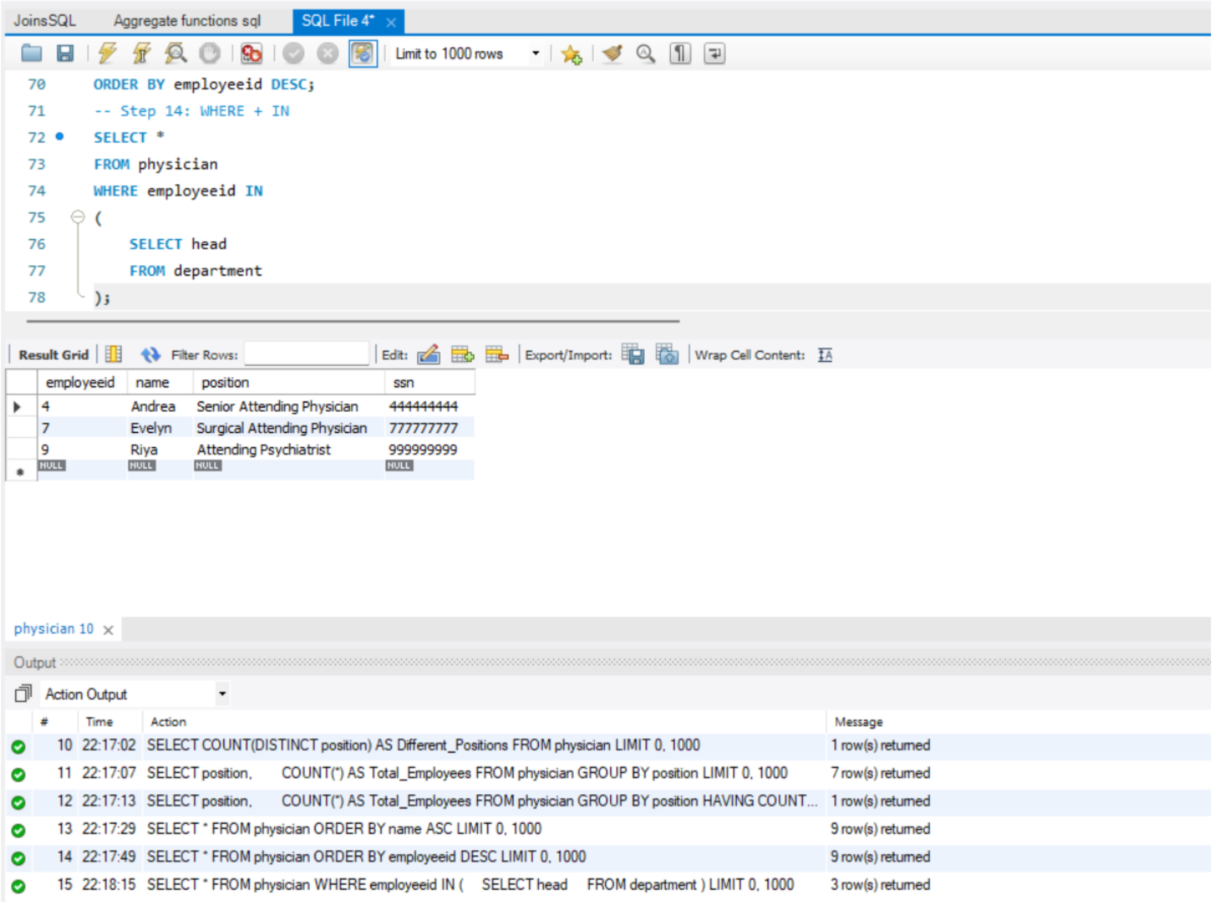
physician 9

Output

Action Output

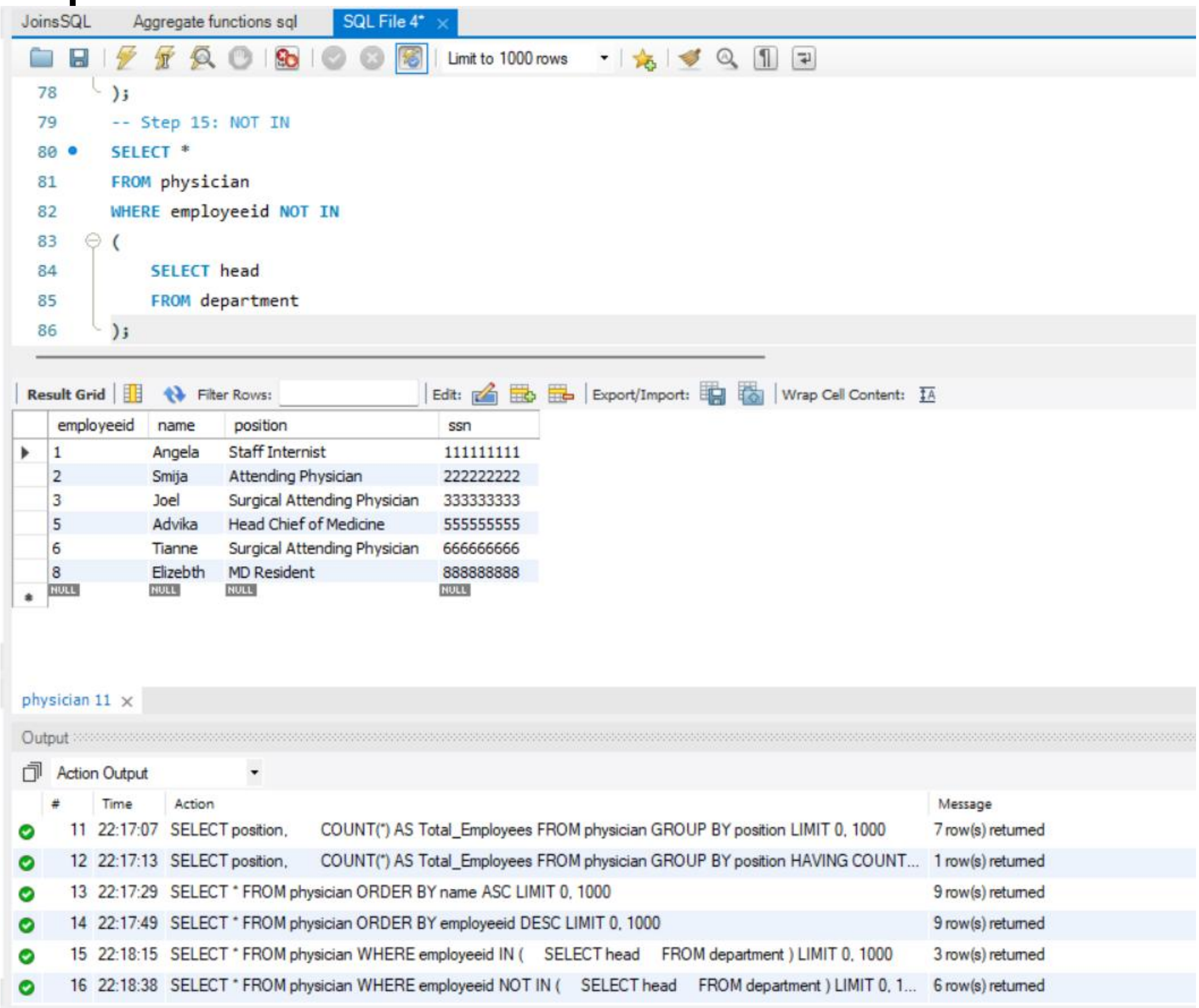
#	Time	Action	Message
9	22:16:59	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
10	22:17:02	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
11	22:17:07	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
12	22:17:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
13	22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned

Step 8 : IN



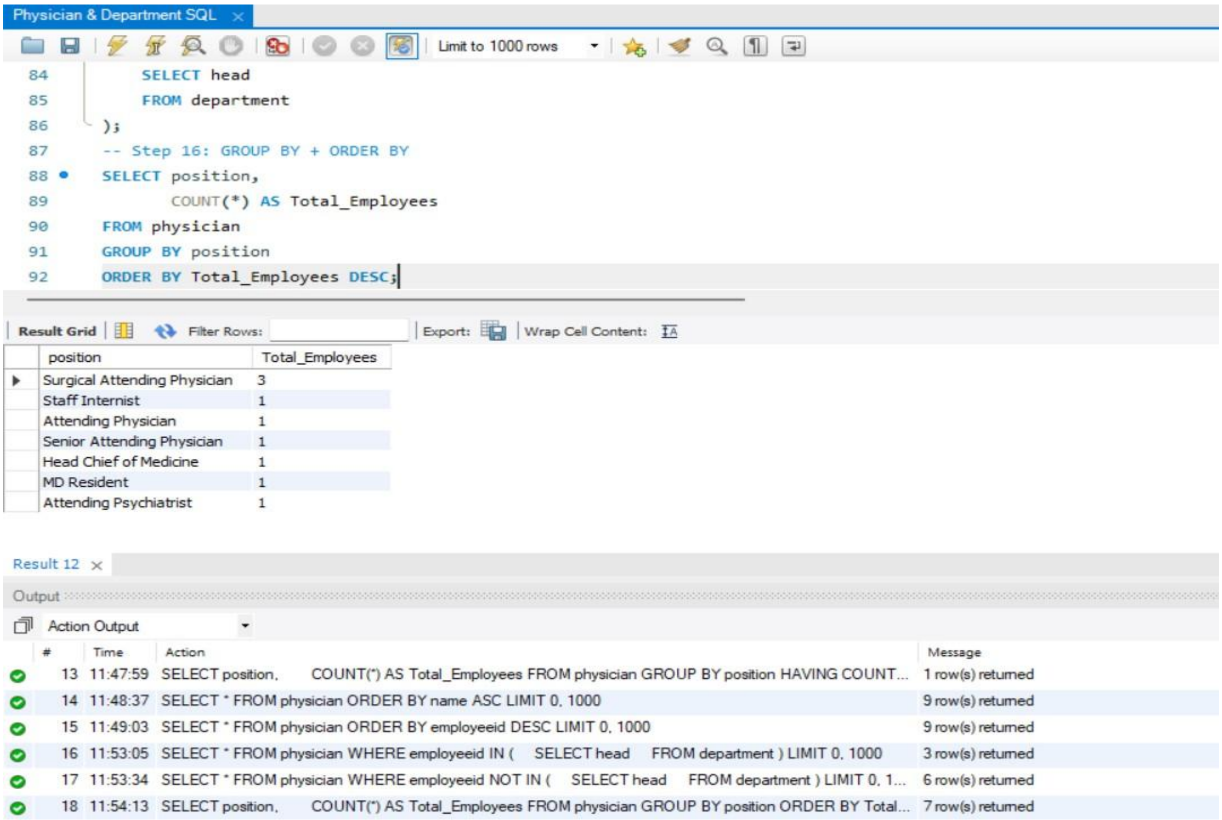
Step 9 : NOT IN

Step 8 : IN



JoinsSQL
Aggregate functions sql
SQL File 4*
Limit to 1000 rows
-- Step 15: NOT IN
SELECT *
FROM physician
WHERE employeeid NOT IN
(
SELECT head
FROM department
);
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employeeid name position ssn
1 Angela Staff Internist 111111111
2 Smija Attending Physician 222222222
3 Joel Surgical Attending Physician 333333333
5 Advika Head Chief of Medicine 555555555
6 Tianne Surgical Attending Physician 666666666
8 Elizebth MD Resident 888888888
NULL NULL NULL NULL
physician 11
Output
Action Output
Time Action Message
11 22:17:07 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000 7 row(s) returned
12 22:17:13 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT... 1 row(s) returned
13 22:17:29 SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000 9 row(s) returned
14 22:17:49 SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000 9 row(s) returned
15 22:18:15 SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000 3 row(s) returned
16 22:18:38 SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1... 6 row(s) returned

Step 10 : GROUP BY + ORDER BY



Step 11: LIKE

Step 10 : GROUP BY + ORDER BY

JoinsSQL | Aggregate functions sql | SQL File 4*

Limit to 1000 rows

```sql
SELECT position,
        COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
ORDER BY Total_Employees DESC;
-- Step 17: WHERE + LIKE
SELECT *
FROM physician
WHERE position LIKE '%Attending%';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employeeid	name	position	ssn
2	Smija	Attending Physician	222222222
3	Joel	Surgical Attending Physician	333333333
4	Andrea	Senior Attending Physician	444444444
6	Tianne	Surgical Attending Physician	666666666
7	Evelyn	Surgical Attending Physician	777777777
9	Riya	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

physician 13

Output

Action Output

#	Time	Action	Message
13	22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
18	22:19:09	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned

Step 12: JOIN + WHERE

JoinsSQL | Aggregate functions sql | SQL File 4*

Limit to 1000 rows

```sql
SELECT *
FROM physician
WHERE position LIKE '%Attending%';
-- Step 18: JOIN + WHERE
SELECT p.name
FROM physician p
JOIN department d
ON p.employeeid = d.head
WHERE d.name = 'Surgery';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name
Evelyn

Result 14

Output

Action Output

#	Time	Action	Message
14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
18	22:19:09	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
19	22:19:30	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned

Step 13: JOIN+ORDER BY

JoinsSQL | Aggregate functions sql | SQL File 4*

Limit to 1000 rows

```sql
ON p.employeeid = d.head
WHERE d.name = 'Surgery';
-- Step 19: JOIN + ORDER BY
SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid
ORDER BY d.name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department	Head_Physician
General Medicine	Andrea
Psychiatry	Riya
Surgery	Evelyn

Result 15

Output

Action Output

#	Time	Action	Message
15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
18	22:19:09	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
19	22:19:30	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
20	22:19:56	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned

Step 14: NOT LIKE

JoinsSQL | Aggregate functions sql | SQL File 4*

Limit to 1000 rows

```sql
FROM department d
JOIN physician p
ON d.head = p.employeeid
ORDER BY d.name;
-- Step 20: WHERE + NOT LIKE
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
-- Step 21: COUNT + HAVING
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employeeid	name	position	ssn
1	Angela	Staff Internist	111111111
2	Smija	Attending Physician	222222222
4	Andrea	Senior Attending Physician	444444444
5	Advika	Head Chief of Medicine	555555555
8	Elizebth	MD Resident	888888888
9	Riya	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

physician 16

Output

Action Output

#	Time	Action	Message
16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
18	22:19:09	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
19	22:19:30	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
20	22:19:56	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
21	22:20:34	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned

Step 15: COUNT+HAVING

Physician & Department SQL

Limit to 1000 rows

```sql
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
-- Step 21: COUNT + HAVING
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

position	Total
Surgical Attending Physician	3

Result 17

Output

Action Output

#	Time	Action	Message
18	11:54:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
19	11:54:38	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMI...	1 row(s) returned

Step 16: EXISTS

Physician & Department SQL

Limit to 1000 rows

```sql
-- Step 22: EXISTS
SELECT *
FROM department d
WHERE EXISTS
(
    SELECT 1
    FROM physician p
    WHERE p.employeeid = d.head
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

department 18

Output

Action Output

	#	Time	Action	Message
✓	19	11:54:38	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
✓	20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓	21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓	22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓	23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMI...	1 row(s) returned
✓	24	12:00:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned

Step 17: MAX

Physician & Department SQL

Limit to 1000 rows

```sql
WHERE EXISTS
(
    SELECT 1
    FROM physician p
    WHERE p.employeeid = d.head
);
-- Step 23: MAX()
SELECT MAX(employeeid) AS Highest_Employee_ID
FROM physician;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Highest_Employee_ID
9

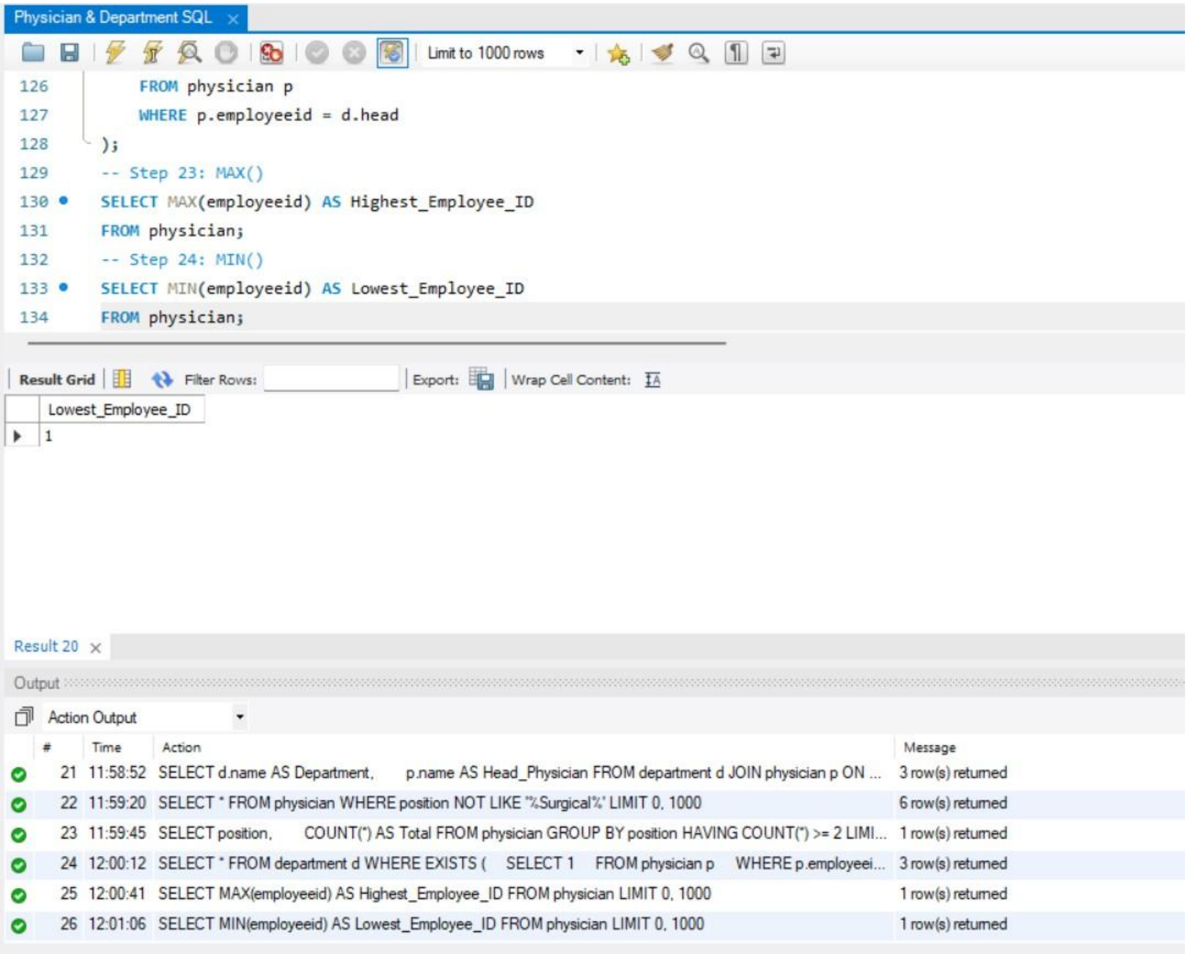
Result 19

Output

Action Output

	#	Time	Action	Message
✓	20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓	21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓	22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓	23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMI...	1 row(s) returned
✓	24	12:00:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned
✓	25	12:00:41	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned

Step 18: MIN


Physician & Department SQL
Limit to 1000 rows
FROM physician p
WHERE p.employeeid = d.head
);
-- Step 23: MAX()
SELECT MAX(employeeid) AS Highest_Employee_ID
FROM physician;
-- Step 24: MIN()
SELECT MIN(employeeid) AS Lowest_Employee_ID
FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Lowest_Employee_ID
1
Result 20
Output
Action Output
Time Action Message
21 11:58:52 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned
22 11:59:20 SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000 6 row(s) returned
23 11:59:45 SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMI... 1 row(s) returned
24 12:00:12 SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei... 3 row(s) returned
25 12:00:41 SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000 1 row(s) returned
26 12:01:06 SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LIMIT 0, 1000 1 row(s) returned